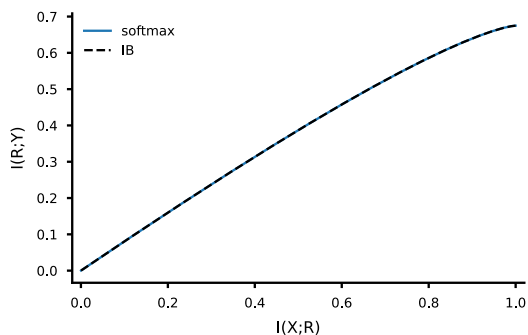
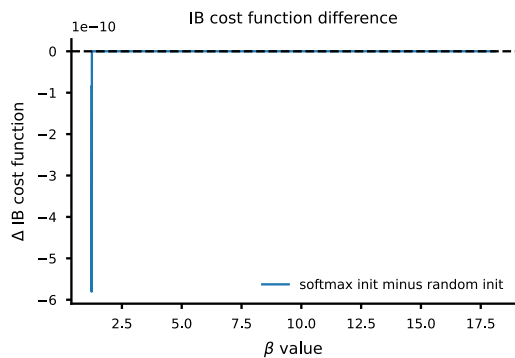


A

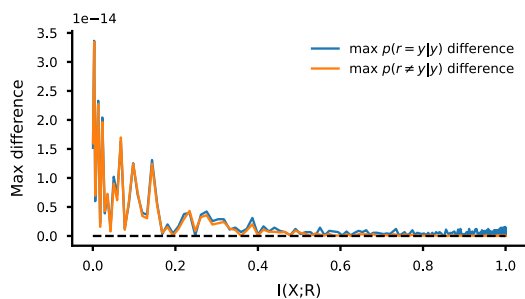
Bead-prediction task
Fully optimal (seven-back) IB bound

**B**

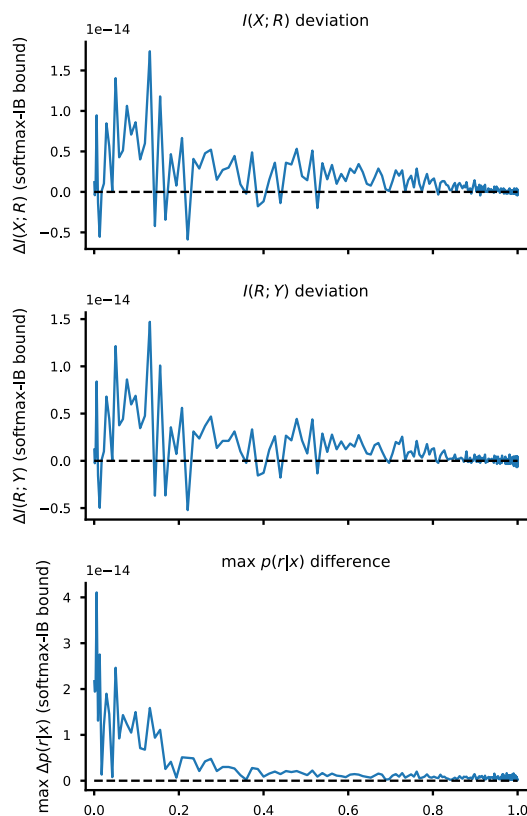
Comparison of softmax initialization to random initialization for IB algorithm (<0 means random is worse)

**C**

Check of $p(r=y|y)$ property that must be true for softmax solution to hold (traces must be 0)

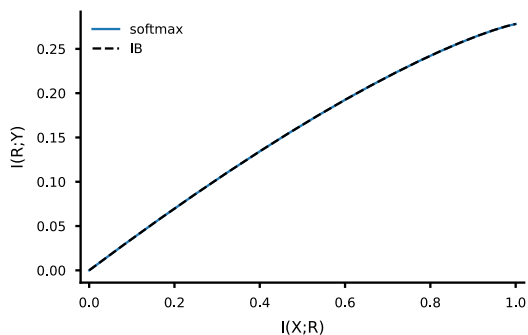
**D**

Comparison of softmax solution to IB algorithm solution
(0 means no difference)

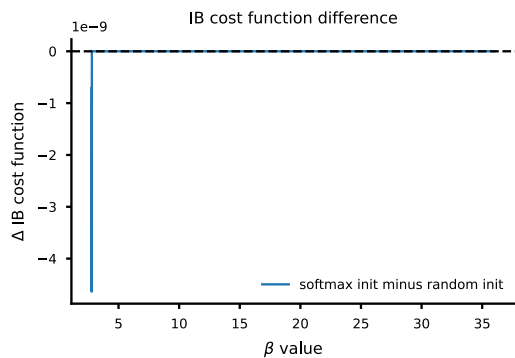


A

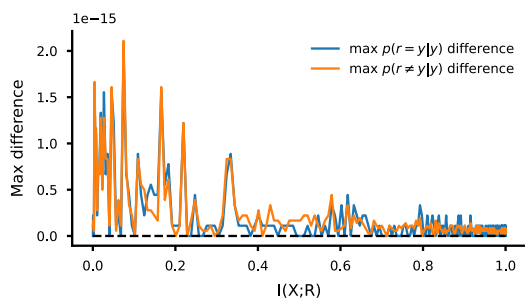
Bead-prediction task
Heuristic (one-back) IB bound

**B**

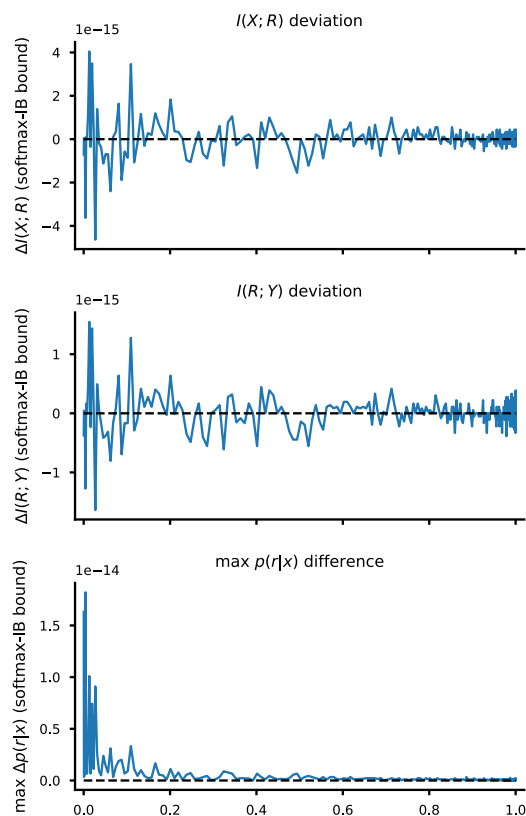
Comparison of softmax initialization to random initialization for IB algorithm (<0 means random is worse)

**C**

Check of $p(r=y|y)$ property that must be true for softmax solution to hold (traces must be 0)

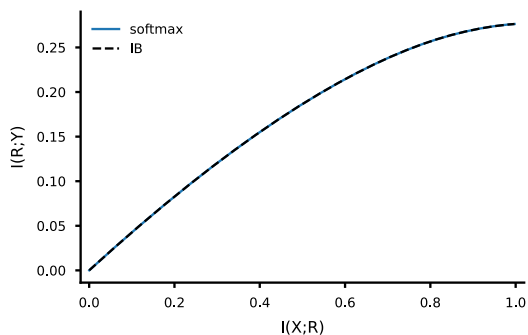
**D**

Comparison of softmax solution to IB algorithm solution
(0 means no difference)

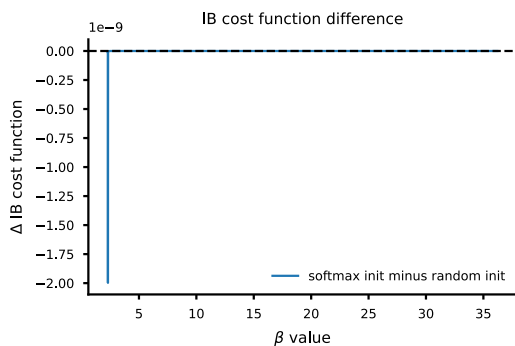


A

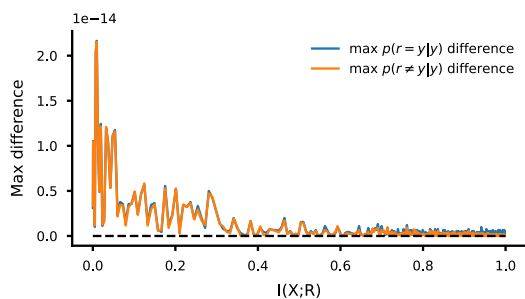
Horse-prediction task
Fully-optimal IB bound

**B**

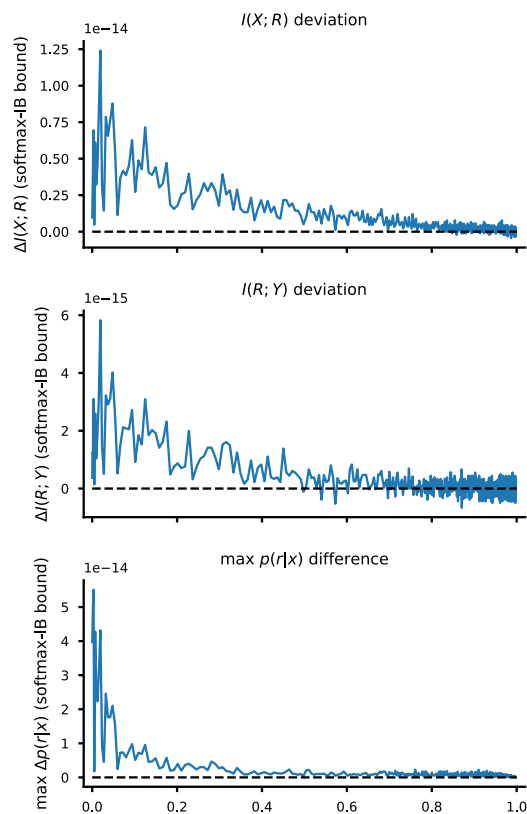
Comparison of softmax initialization to random initialization for IB algorithm (<0 means random is worse)

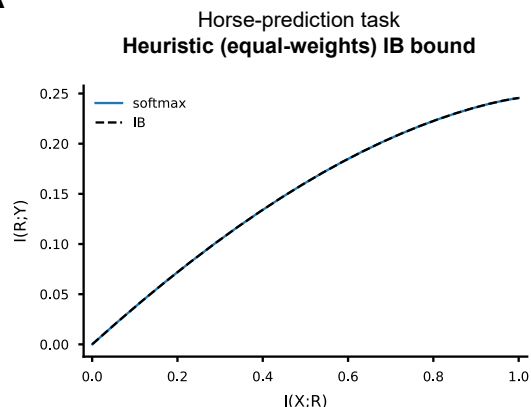
**C**

Check of $p(r=y|y)$ property that must be true for softmax solution to hold (traces must be 0)

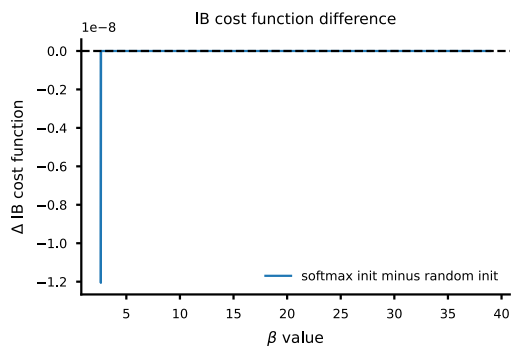
**D**

Comparison of softmax solution to IB algorithm solution
(0 means no difference)

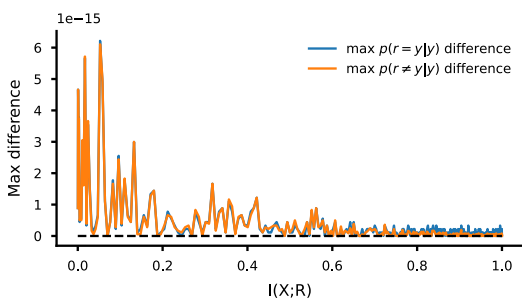


A**B**

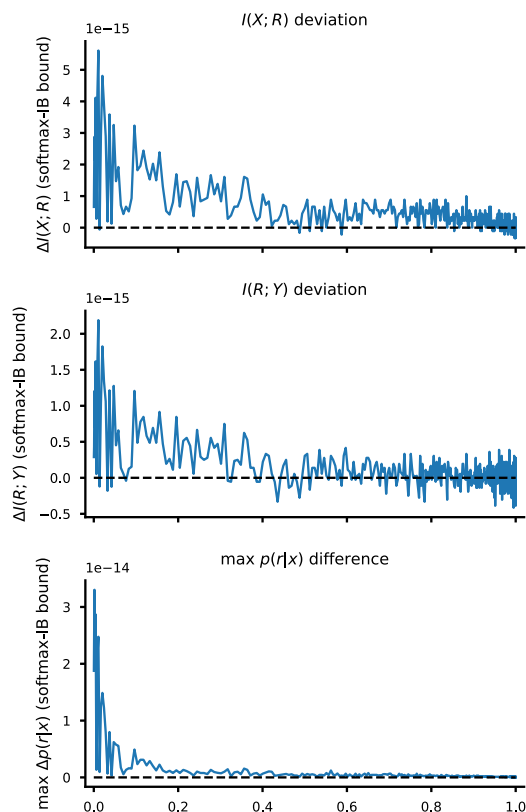
Comparison of softmax initialization to random initialization for IB algorithm (<0 means random is worse)

**C**

Check of $p(r=y|y)$ property that must be true for softmax solution to hold (traces must be 0)

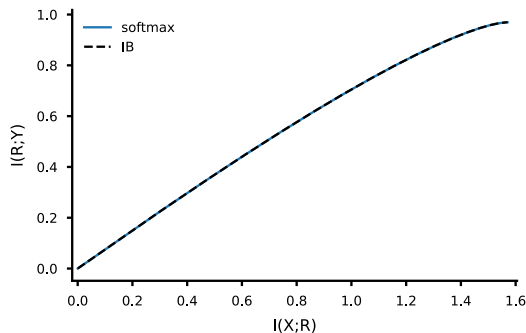
**D**

Comparison of softmax solution to IB algorithm solution
(0 means no difference)

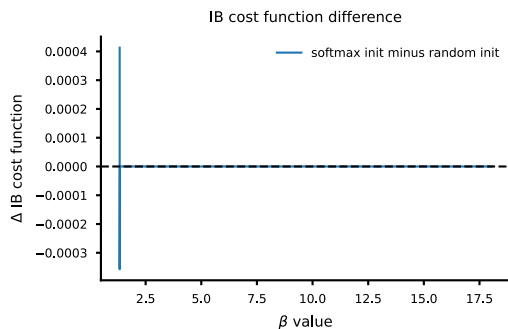


A

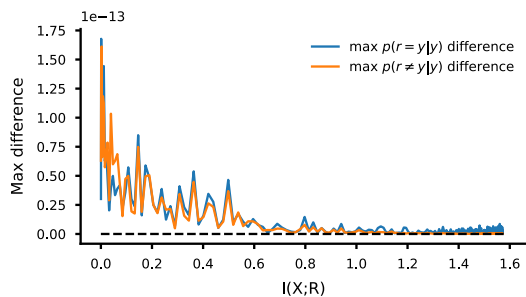
Three jar bead-prediction task
Fully-optimal (seven-back) IB bound

**B**

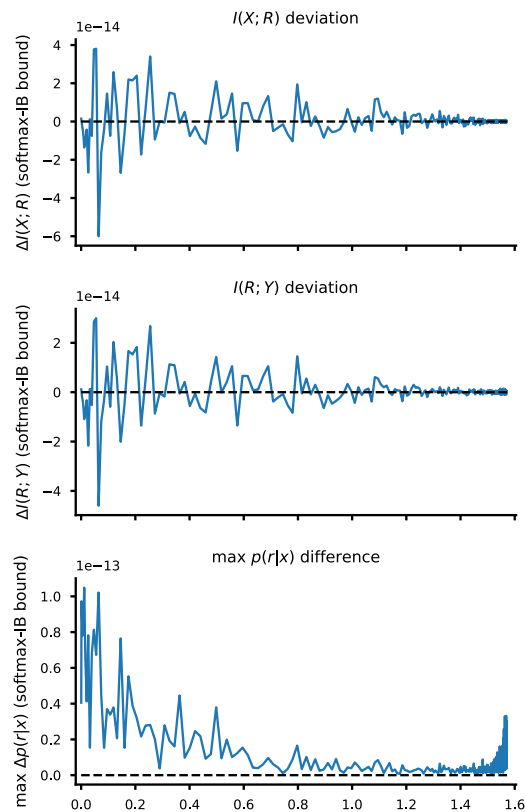
Comparison of softmax initialization to random initialization for IB algorithm (<0 means random is worse)

**C**

Check of $p(r=y|y)$ property that must be true for softmax solution to hold (traces must be 0)

**D**

Comparison of softmax solution to IB algorithm solution
(0 means no difference)

**E**

Hidden Markov model parameters

Transition matrix, **H**
P(jar trial $t+1$ | jar trial t)

$$\begin{bmatrix} 0.98 & 0.01 & 0.01 \\ 0.01 & 0.98 & 0.01 \\ 0.01 & 0.01 & 0.98 \end{bmatrix}$$

Emission matrix, **E**
P(bead trial t | jar trial t)

$$\begin{bmatrix} 0.70 & 0.15 & 0.15 \\ 0.15 & 0.70 & 0.15 \\ 0.15 & 0.15 & 0.70 \end{bmatrix}$$